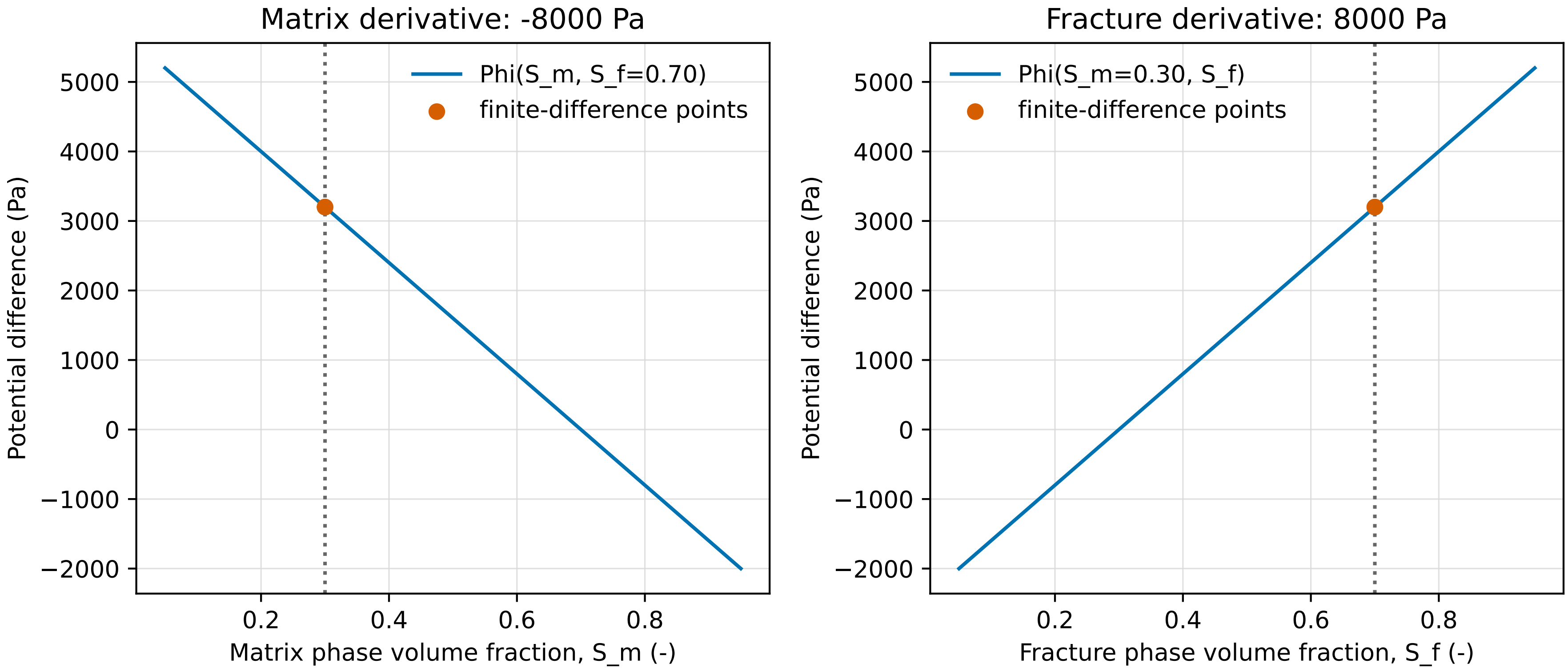

Reference

Phi=p_m-(a+bS_m)-p_f+(a+bS_f), b=8000 Pa.
Expected dPhi/dS_m=-8000 Pa; dPhi/dS_f=+8000 Pa.

This run

Central differences: -8000.000000 Pa and 8000.000000 Pa.
Both relative errors are below 1e-10.

Physical picture

Increasing matrix saturation raises matrix capillary pressure and lowers Phi.
Increasing fracture saturation raises fracture capillary pressure and raises Phi.
This is an algebraic contract, not a runtime field-mapping test.